

Technique 1: dx-dt Transform

MP

Teach You Physics Hand In Hand

April 2026

1 SOLUTION to Problem in Gravitation Post

- Thanks to Schoolbabyboy for proposing this question.
- Level of difficulty of the problem: **7/10**

A-(ai): This is simply the definition of $\vec{g}(r)$. Just show it in mathematical calculation.

$$-\nabla V(r) = -\frac{\partial V}{\partial r} = -\frac{\partial}{\partial r} \frac{GM}{r} = \frac{GM}{r^2} = \vec{g}(r) \quad [1 \text{ point}]$$

A-(aai): Force: $\vec{F} = -\frac{km}{r^p} \hat{r} = -\nabla(mV)$. Hence,

$$\frac{dV}{dr} = \frac{k}{r^p} \Rightarrow V(r) = -\frac{k}{(p-1)r^{p-1}} \quad [1 \text{ point}]$$

A-(aiii): Initial $r_0 = y_0$, $\dot{r}(0) = 0$. Energy conservation gives: [2 points]

$$\frac{1}{2}m\dot{r}^2 - \frac{km}{(p-1)r^{p-1}} = -\frac{km}{(p-1)y_0^{p-1}} \Rightarrow \dot{r} = -\sqrt{\frac{2k}{p-1} \left(\frac{1}{r^{p-1}} - \frac{1}{r_0^{p-1}} \right)}$$

Time to hit the origin:

$$T = \int_{y_0}^0 \frac{dr}{\dot{r}} = \int_0^{y_0} \frac{dr}{\sqrt{\frac{2k}{p-1} (r^{-(p-1)} - y_0^{-(p-1)})}} \quad [1 \text{ point}]$$

B-(bi): Final speed v_∞ as $r \rightarrow \infty$:

$$\frac{1}{2}v_\infty^2 = \frac{1}{2}v_0^2 - \frac{k}{(p-1)r_0^{p-1}} \Rightarrow v_\infty = \sqrt{v_0^2 - \frac{2k}{(p-1)r_0^{p-1}}} \quad [1 \text{ point}]$$

Unbounded motion requires $E > 0$ ($v_\infty > 0$):

$$v_0 > \sqrt{\frac{2k}{(p-1)r_0^{p-1}}} \quad [1 \text{ point}]$$

B-(bii): Angular momentum per mass: $h = v_0 x_0$. [1 point] Let $u = 1/r$, $\dot{r} = -h \frac{du}{d\theta}$ we get:

$$E^* = \frac{1}{2} \dot{r}^2 + \frac{h^2}{2r^2} - \frac{k}{(p-1)r^{p-1}} \Rightarrow \left(\frac{du}{d\theta} \right)^2 = \frac{2E^*}{h^2} - u^2 + \frac{2k}{(p-1)h^2} u^{p-1} \quad [3 \text{ points}]$$

Define E^* to be energy per mass. Hence:

$$\theta\left(\frac{1}{r}\right) = \theta_0 + \int_{1/r}^{1/r_0} \frac{d\left(\frac{1}{r}\right)}{\sqrt{\frac{2E^*}{v_0^2 x_0^2} - \left(\frac{1}{r}\right)^2 + \frac{2k}{(p-1)v_0^2 x_0^2} \left(\frac{1}{r}\right)^{p-1}}} \quad [2 \text{ points}]$$

C-(ci): Given $\frac{\partial V}{\partial t} + \alpha V = 0 \Rightarrow V(r, t) = V(r, 0)e^{-\alpha t}$. Take Laplacian:

$$\nabla^2 V(r, t) = e^{-\alpha t} \nabla^2 V(r, 0) \propto e^{-\alpha t} \rho(r, 0) \Rightarrow \rho(r, t) = e^{-\alpha t} \rho(r, 0) \quad [1 \text{ point}]$$

C-(cii): We can see that:

$$V_{\text{eff}}(r, t) \approx V_{\text{eff}}(r_0, t) + \frac{1}{2} \kappa(t) (r - r_0)^2$$

where $\kappa(t) = \delta \omega(t)^2$ and $V_{\text{eff}}(r_0, t) = -\frac{k}{(p-1)r_0^{p-1}} e^{-\alpha t} + \frac{\hbar^2 L^2}{2\delta r_0^2}$

$$\Rightarrow \kappa(t) = -\frac{kp}{r_0^{p+1}} e^{-\alpha t} + \frac{3\hbar^2 L^2}{\delta r_0^4} \quad [2 \text{ points}]$$

Substitute into Schrödinger Equation:

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2\delta} \frac{\partial^2 \Psi}{\partial x^2} + \left(V(r, t) + \frac{\hbar^2 L^2}{2\delta r^2} \right) \Psi$$

$$\Rightarrow i\hbar \dot{A} - i\hbar \frac{1}{2} \sigma \dot{B} x^2 A = -\frac{\hbar^2}{2\delta} A [-\sigma B + (\sigma B)^2 x^2] + \left[V_{\text{eff}}(r_0, t) + \frac{1}{2} \kappa x^2 \right] A \quad [2 \text{ points}]$$

$$\Rightarrow \dot{B} = -i \frac{\hbar \sigma}{\delta} B^2 + i \frac{1}{\hbar \sigma} \left(-\frac{kp}{r_0^{p+1}} e^{-\alpha t} + \frac{3\hbar^2 L^2}{\delta r_0^4} \right) \quad [1 \text{ point}]$$

$$\Rightarrow \dot{A} = -i \frac{\hbar \sigma}{2\delta} B A - i \frac{1}{\hbar} \left(-\frac{k}{(p-1)r_0^{p-1}} e^{-\alpha t} + \frac{\hbar^2 L^2}{2\delta r_0^2} \right) A \quad [1 \text{ point}]$$

2 dx-dt Transform

When we are asked to find the time taken of certain motions that are non-linear, given the equation of motion, we can use dx-dt transformation to find the time taken. The dx-dt transformation is just formula (1) and (2).

$$\ddot{x} = \frac{d\dot{x}}{dt} = \frac{dx}{dt} \frac{d\dot{x}}{dx} = \dot{x} \frac{d\dot{x}}{dx} \quad (1)$$

It expresses acceleration in terms of velocity, which is a function of displacement. To find the time taken, we can simply write:

$$T = \int dt = \int \frac{dx}{\dot{x}} \quad (2)$$

If you can directly see the expression of $\dot{x}$ from energy conservation (we will cover it in later parts), you can skip the derivative part.

3 Example

Consider a strange oscillatory motion described by:

$$m\ddot{x} = -\alpha x^2 \quad (3)$$

Substituting (1) into (3) yields:

$$m\dot{x} \frac{d\dot{x}}{dx} = -\alpha x^2 \quad (4)$$

Define x_0 to be the amplitude. Integration gives:

$$\frac{1}{2}m\dot{x}^2 + \frac{1}{3}\alpha x^3 = \frac{1}{3}\alpha x_0^3 \quad (5)$$

NB!!! We derived the form of energy conservation from fundamental derivatives. Therefore, the velocity (a function of displacement) can be expressed as:

$$\dot{x} = \sqrt{\frac{2\alpha}{3m}(x_0^3 - x^3)} \quad (6)$$

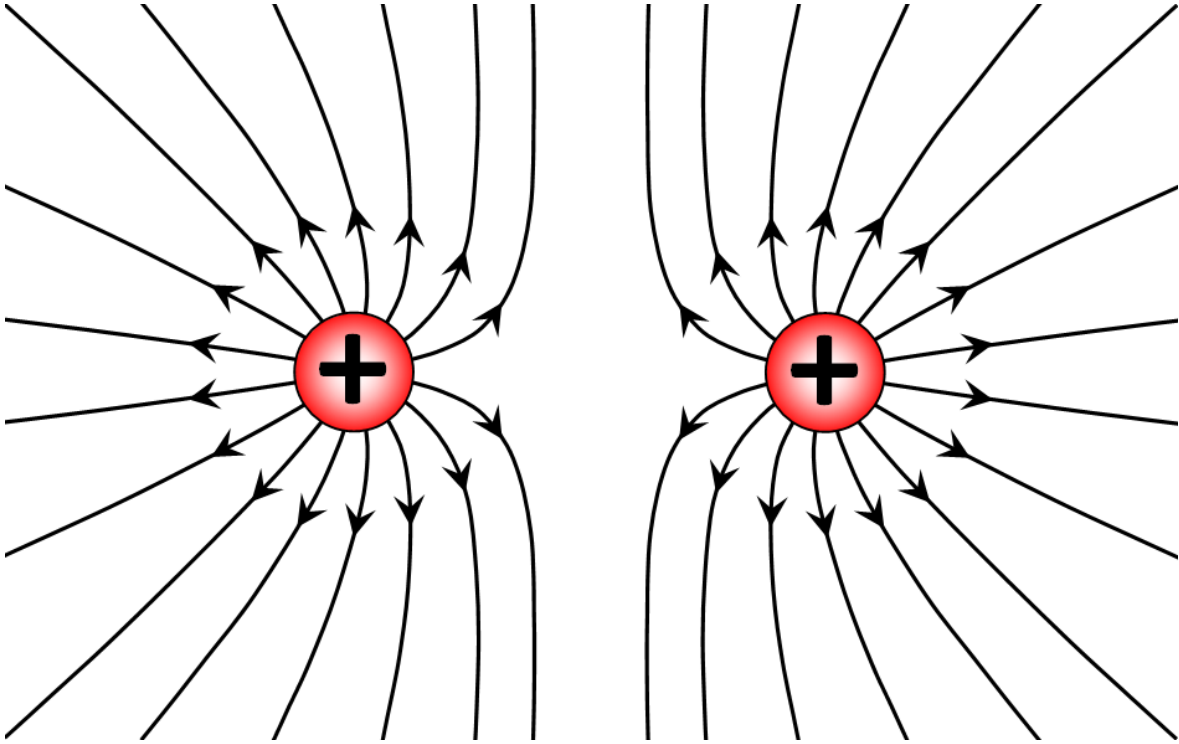
Since the motion is a harmonic motion, period can be expressed as 2 times the time taken for traveling from peak to peak. Substituting (6) into (2) yields:

$$T = \int dt = \int \frac{dx}{\dot{x}} = 2 \int_{-x_0}^{x_0} \frac{dx}{\sqrt{\frac{2\alpha}{3m}(x_0^3 - x^3)}} \quad (7)$$

And the problem is solved. You will be provided with an integration formula to solve T in the question. Of course if you can see the energy conservation directly, you can directly write energy conservation which is (5) and solve.

4 Problem

A good problem involving dx-dt transformation is the problem in Gravitation Post by Schoolbabyboy. Please make sure that you revise it together with the solution provided in this post.



[20 points] Consider 2 classical point charges of same mass m and same charge of $+q$. Initially, they are compressed together with initial separation of r_0 . The system is then released from rest. Denote the position of the charges be $r_1(t)$ and $r_2(t)$. Define their separation be $r(t) = r_2(t) - r_1(t)$. It is given that the motion is one-dimensional. The permittivity of vacuum is ϵ_0 .

a) [4 points] Derive a differential equation for $r(t)$.

b) [16 points] Find the time required for the charges to push away themselves from one another and attain a separation of L . Hint: try substitution of $r(t) = \frac{1}{u(t)}$

You may use the fact that:

$$\int \frac{du}{u^2 \sqrt{a-u}} = -\frac{\sqrt{a-u}}{au} - \frac{1}{2a^{3/2}} \ln \left| \frac{\sqrt{a} + \sqrt{a-u}}{\sqrt{a} - \sqrt{a-u}} \right| + C$$

The solution will be uploaded in the next post and MP love you all :)